

**MATHEMATICS Compulsory Part  
PAPER 2**

11:30 am – 12:45 pm (1¼ hours)

**INSTRUCTIONS**

1. Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first stick a barcode label and insert the information required in the spaces provided. No extra time will be given for sticking on the barcode label after the ‘Time is up’ announcement.
2. When told to open this book, you should check that all the questions are there. Look for the words ‘**END OF PAPER**’ after the last question.
3. All questions carry equal marks.
4. **ANSWER ALL QUESTIONS.** You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answers clearly; otherwise you will lose marks if the answers cannot be captured.
5. You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.
6. No marks will be deducted for wrong answers.

There are 30 questions in Section A and 15 questions in Section B.  
The diagrams in this paper are not necessarily drawn to scale.  
Choose the best answer for each question.

Section A

1.  $(x+3y)^2 - (x-3y)^2 =$

A.  $2x^2$  .

B.  $6xy$  .

C.  $12xy$  .

D.  $2x^2 + 18y^2$  .

2.  $\frac{(2\alpha)^3}{(4\alpha^{-5})^{-1}} =$

A.  $2\alpha^8$  .

B.  $32\alpha^8$  .

C.  $\frac{2}{\alpha^2}$  .

D.  $\frac{32}{\alpha^2}$  .

3. If  $k = \frac{5}{2m} + n$  , then  $m =$

A.  $\frac{5}{2(k-n)}$  .

B.  $\frac{5}{2(n-k)}$  .

C.  $\frac{2(k-n)}{5}$  .

D.  $\frac{2(n-k)}{5}$  .

4.  $\sqrt{333} =$

- A. 18 (correct to the nearest integer).
- B. 18.24 (correct to 2 decimal places).
- C. 18.248 (correct to 3 significant figures).
- D. 18.2482 (correct to 4 decimal places).

5. The price of 2 apples and 3 lemons is \$38 . If the price of 3 apples and 2 lemons is \$47 , then the price of 4 apples and 7 lemons is

- A. \$78 .
- B. \$80 .
- C. \$82 .
- D. \$84 .

6. If  $a$  ,  $b$  and  $c$  are non-zero constants such that  $4x^2 + 2ax + 3a \equiv x(4x + b) + 2c$  , then  $a : b : c =$

- A. 2 : 4 : 3 .
- B. 3 : 4 : 2 .
- C. 4 : 6 : 3 .
- D. 6 : 4 : 3 .

7. Let  $m$  be a constant. Solve the equation  $x^2 - 3x = (m - 1)^2 - 3(m - 1)$  .

- A.  $x = m - 1$  or  $x = m - 4$
- B.  $x = m - 1$  or  $x = 4 - m$
- C.  $x = 1 - m$  or  $x = m - 4$
- D.  $x = 1 - m$  or  $x = 4 - m$

8. Let  $g(x) = (x+1)(x+a)$ , where  $a$  is a constant. If  $g(1) = g(2)$ , then  $g(a) =$

- A.  $-4$ .
- B.  $0$ .
- C.  $12$ .
- D.  $24$ .

9. Let  $f(x) = x^3 + kx^2 + 5x + 10$ , where  $k$  is a constant. If  $f(x)$  is divisible by  $x+k$ , find the remainder when  $f(x)$  is divided by  $x+1$ .

- A.  $-2$
- B.  $2$
- C.  $6$
- D.  $18$

10. The solution of  $\frac{1-x}{2} \geq 4$  or  $7+5x \leq -3$  is

- A.  $x \leq -7$ .
- B.  $x \leq -2$ .
- C.  $-7 \leq x \leq -2$ .
- D.  $x \leq -7$  or  $x \geq -2$ .

11. In a school, 40% of the students are girls and  $\beta\%$  of the girls are foreign students. It is given that 30% of the boys in the school are foreign students. In the school, the number of foreign students and the number of girls are equal. Find  $\beta$ .

- A. 20
- B. 45
- C. 55
- D. 80

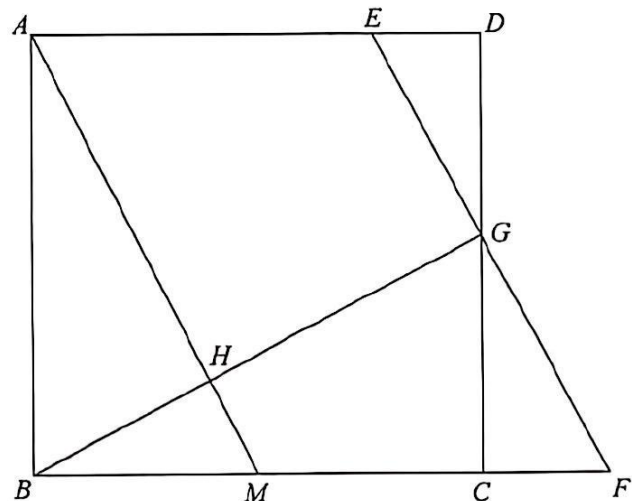
12. A car travels at an average speed of 60 km/h for 18 minutes and then the car travels at an average speed of 40 km/h for 27 minutes. The average speed of the car for the whole journey is
- A. 48 km/h .  
B. 50 km/h .  
C. 52 km/h .  
D. 54 km/h .
13. It is given that  $z$  varies directly as the square of  $x$  and inversely as  $y$ . If  $x$  is increased by 20% and  $y$  is decreased by 20% , then  $z$
- A. is increased by 20% .  
B. is decreased by 20% .  
C. is increased by 80% .  
D. is decreased by 80% .
14. Which of the following statements about the graph of  $y = 2(6 - x)^2 - 7$  is true?
- A. The graph opens upwards.  
B. The graph does not cut the  $x$ -axis.  
C. The  $y$ -intercept of the graph is  $-7$  .  
D. The graph passes through the point  $(-6, -7)$  .
15. If the arc length and the area of a sector are  $8\pi$  cm and  $80\pi$  cm<sup>2</sup> respectively, then the angle of the sector is
- A.  $36^\circ$  .  
B.  $45^\circ$  .  
C.  $60^\circ$  .  
D.  $72^\circ$  .

16. The ratio of the height of a right circular cylinder to the height of a right circular cone is  $32:15$  while the ratio of the volume of the circular cylinder to the volume of the circular cone is  $10:9$ . If the base radius of the circular cylinder is  $25\text{ cm}$ , then the base radius of the circular cone is

- A.  $20\text{ cm}$ .
- B.  $24\text{ cm}$ .
- C.  $48\text{ cm}$ .
- D.  $60\text{ cm}$ .

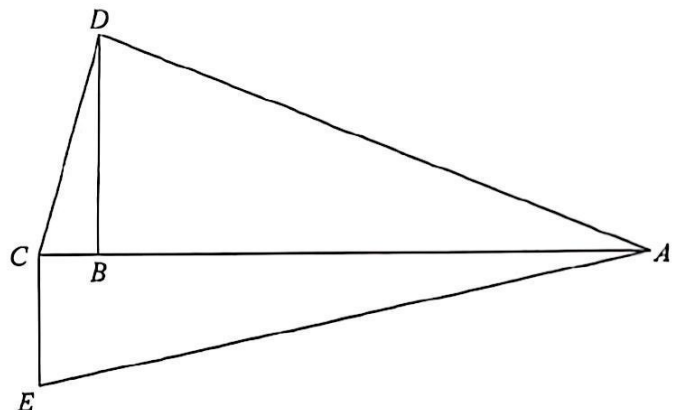
17. In the figure,  $ABCD$  is a square. Let  $M$  be the mid-point of  $BC$ .  $E$  is a point lying on  $AD$  such that  $AE:ED = 3:1$ .  $F$  is a point lying on  $BC$  produced such that  $EF \parallel AM$ .  $CD$  and  $EF$  intersect at the point  $G$  while  $AM$  and  $BG$  intersect at the point  $H$ . If the area of  $\triangle BMH$  is  $4\text{ cm}^2$ , then the area of the trapezium  $AEGH$  is

- A.  $12\text{ cm}^2$ .
- B.  $33\text{ cm}^2$ .
- C.  $39\text{ cm}^2$ .
- D.  $45\text{ cm}^2$ .



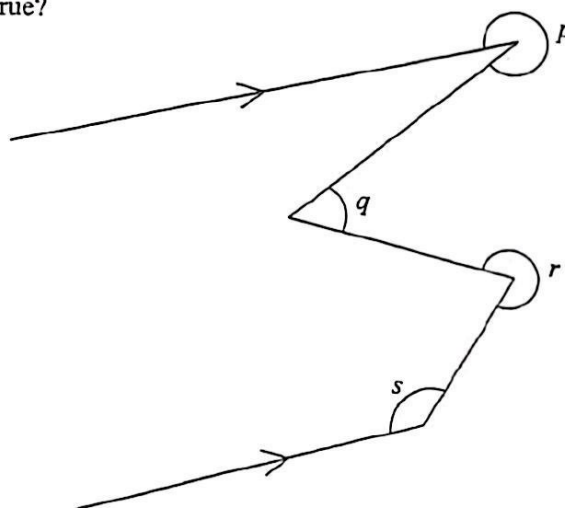
18. In the figure,  $ABC$  is a straight line. It is given that  $AD = 37\text{ cm}$ ,  $BC = 5\text{ cm}$ ,  $BD = 12\text{ cm}$ ,  $CD = 13\text{ cm}$  and  $CE = 9\text{ cm}$ . If  $\angle ACE = 90^\circ$ , find the perimeter of the quadrilateral  $ADCE$ .

- A.  $76\text{ cm}$
- B.  $90\text{ cm}$
- C.  $100\text{ cm}$
- D.  $180\text{ cm}$



19. According to the figure, which of the following must be true?

- A.  $p + q - r = 90^\circ$
- B.  $p - r + s = 180^\circ$
- C.  $p + q - r + s = 270^\circ$
- D.  $p + q + r - s = 540^\circ$



20. If the sum of the interior angles of a regular polygon is  $900^\circ$ , which of the following is/are true?

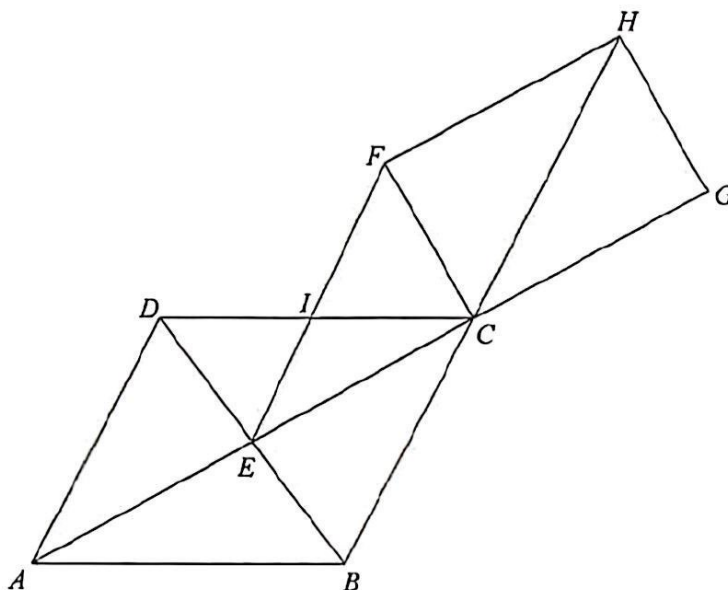
- I. The number of diagonals of the polygon is 7.
- II. The number of folds of rotational symmetry of the polygon is 7.
- III. The number of axes of reflectional symmetry of the polygon is 7.

- A. I only
- B. II only
- C. I and III only
- D. II and III only

21. In the figure,  $ABCD$  is a rhombus. Denote the point of intersection of  $AC$  and  $BD$  by  $E$ . Let  $F$  be a point such that  $BH \parallel EF$  and  $CFHG$  is a rectangle, where  $G$  and  $H$  are points lying on  $AC$  produced and  $BC$  produced respectively. Denote the point of intersection of  $CD$  and  $EF$  by  $I$ . Which of the following must be true?

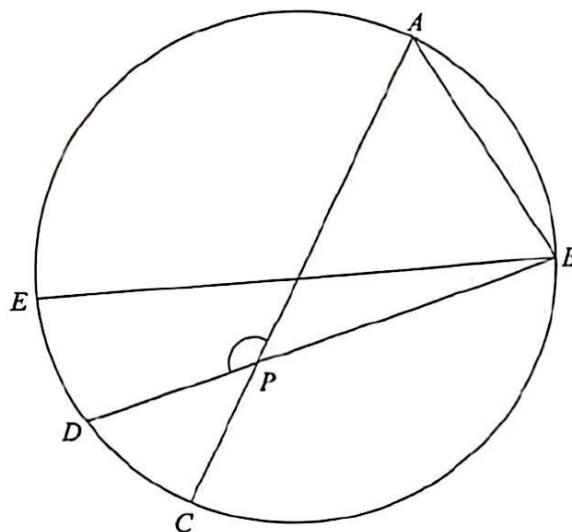
- I.  $CI = FI$
- II.  $\angle ABE = \angle GCH$
- III.  $\triangle ADE \cong \triangle HCF$

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III



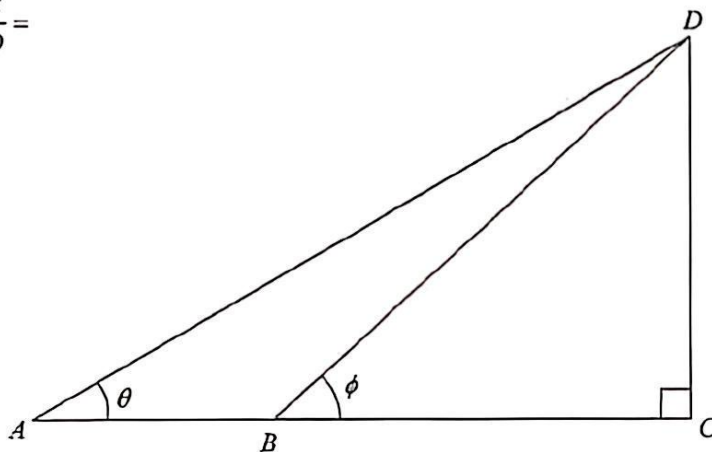
22. In the figure,  $ABCDE$  is a circle.  $AC$  and  $BE$  are diameters of the circle. Let  $P$  be the point of intersection of  $AC$  and  $BD$ . If  $\angle ABE = 46^\circ$  and  $\angle DBE = 16^\circ$ , then  $\angle APD =$

- A.  $104^\circ$  .  
 B.  $108^\circ$  .  
 C.  $120^\circ$  .  
 D.  $135^\circ$  .



23. In the figure,  $ABC$  is a straight line.  $\frac{BC}{AD} =$

- A.  $\frac{\sin \theta}{\tan \phi}$  .  
 B.  $\frac{\tan \phi}{\sin \theta}$  .  
 C.  $\frac{\cos \theta}{\tan \phi}$  .  
 D.  $\frac{\tan \phi}{\cos \theta}$  .



24. The coordinates of the point  $U$  are  $(-3, -8)$ .  $U$  is rotated anticlockwise about the origin through  $90^\circ$  to the point  $V$ .  $V$  is then reflected with respect to the straight line  $x = 2$  to the point  $W$ . Find the  $x$ -coordinate of  $W$ .

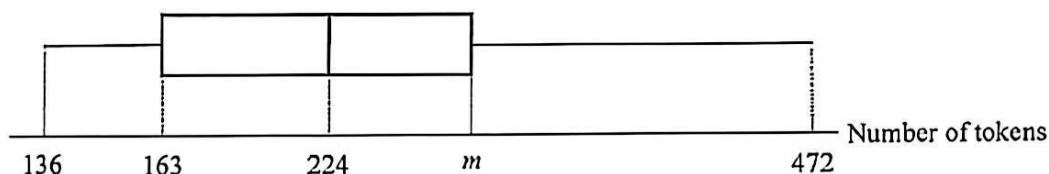
- A.  $-4$   
 B.  $-3$   
 C.  $7$   
 D.  $12$

25. The coordinates of the points  $A$  and  $B$  are  $(-3, 1)$  and  $(-7, -5)$  respectively. If  $P$  is a point lying on the straight line  $x - y + 13 = 0$  such that  $AP = PB$ , then the  $y$ -coordinate of  $P$  is
- A.  $-11$  .  
B.  $-2$  .  
C.  $2$  .  
D.  $11$  .
26. Find the constant  $k$  such that the straight lines  $6x - 8y = 7k$  and  $kx + 12y = 5$  do not intersect with each other.
- A.  $-16$   
B.  $-9$   
C.  $9$   
D.  $16$
27. Denote the circle  $3x^2 + 3y^2 - 6x + 12y - 4 = 0$  by  $C$ . Which of the following are true?
- I. The origin lies inside  $C$  .  
II. The circumference of  $C$  is less than  $16$  .  
III. The perpendicular distance from the centre of  $C$  to the  $x$ -axis is  $2$  .
- A. I and II only  
B. I and III only  
C. II and III only  
D. I, II and III

28. Two numbers are randomly drawn at the same time from six cards numbered 1, 2, 3, 4, 5 and 6 respectively. Find the probability that the product of the numbers drawn is not less than 12.

- A.  $\frac{1}{3}$   
 B.  $\frac{2}{3}$   
 C.  $\frac{7}{15}$   
 D.  $\frac{8}{15}$

29. The box-and-whisker diagram below shows the distribution of the numbers of tokens got by a group of children in a game. If the range of the distribution is the triple of its inter-quartile range, find  $m$ .



- A. 248  
 B. 275  
 C. 336  
 D. 360

30. Consider the following positive integers:

5      5      5      6      9      9      11      13       $m$        $n$

Let  $p$ ,  $q$  and  $r$  be the standard deviation, the mode and the median of the above positive integers respectively. If the mean of the above positive integers is 7, which of the following must be true?

- I.  $p > 3$   
 II.  $q = 5$   
 III.  $r < 7$

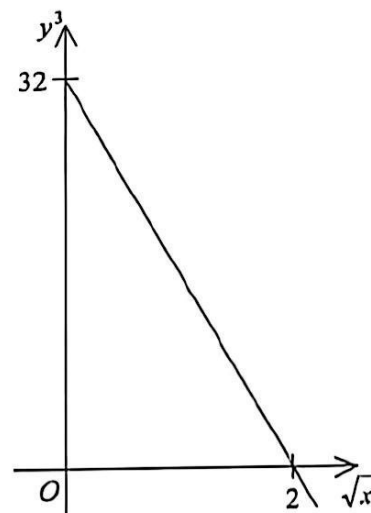
- A. I and II only  
 B. I and III only  
 C. II and III only  
 D. I, II and III

Section B

31. The H.C.F. of  $u^2v^3w$ ,  $u^3vw^2$  and  $u^2v^3w^4$  is
- A.  $uvw$  .
- B.  $u^2vw$  .
- C.  $u^2v^3w^4$  .
- D.  $u^3v^3w^4$  .
32.  $AF00000000BC_{16} =$
- A.  $175 \times 16^{11} + 188$  .
- B.  $192 \times 16^{11} + 205$  .
- C.  $175 \times 16^{12} + 188$  .
- D.  $192 \times 16^{12} + 205$  .
33. If  $x = \log_2 y - 2$  and  $(\log_2 y)^2 = 5 \log_2 y + x - 7$ , then  $y =$
- A. 1 .
- B. 8 .
- C. 1 or 3 .
- D. 3 or 8 .

34. The graph in the figure shows the linear relation between  $y^3$  and  $\sqrt{x}$ . If  $x = 36$ , then  $y =$

- A.  $-64$ .
- B.  $-16$ .
- C.  $-8$ .
- D.  $-4$ .



35. Let  $z = (a-5)i + \frac{(a+2)i}{2+i}$ . If  $a$  and  $z$  are real numbers, then  $a - z =$

- A. 2.
- B. 3.
- C. 4.
- D. 5.

36. The sum of the first  $n$  terms of a sequence is  $n(2n+3)$ . Which of the following are true?

- I. 14 is a term of the sequence.
- II. The  $n$ th term of the sequence is  $4n+1$ .
- III. The sequence is an arithmetic sequence.

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

37. Consider the following system of inequalities:

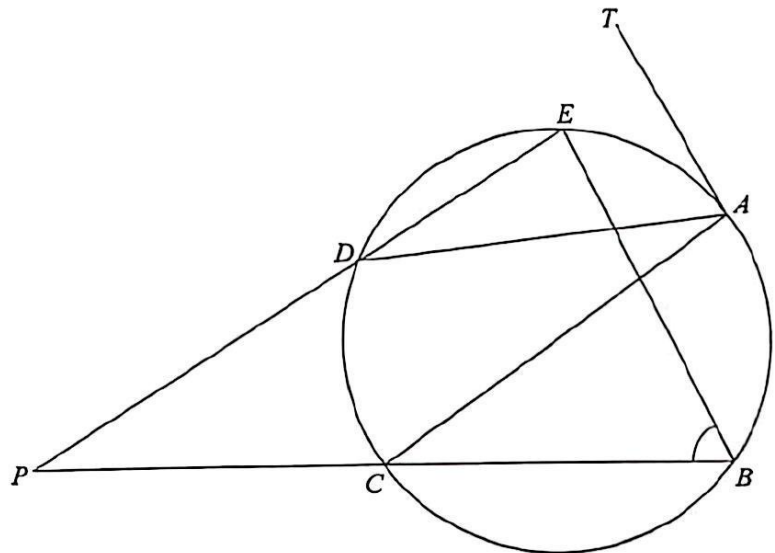
$$\begin{cases} x-2y \leq 1 \\ x+4y \leq 13 \\ 2x-y \geq -1 \end{cases}$$

Let  $R$  be the region which represents the solution of the above system of inequalities. Find the constant  $c$  such that the least value of  $5x-2y+c$  is 22, where  $(x, y)$  is a point lying in  $R$ .

- A. 1
- B. 23
- C. 25
- D. 29

38. In the figure,  $TA$  is the tangent to the circle  $ABCDE$  at the point  $A$ .  $BC$  produced and  $ED$  produced meet at the point  $P$ . If  $\angle ACB = 43^\circ$ ,  $\angle DAT = 55^\circ$  and  $\angle CPD = 29^\circ$ , then  $\angle CBE =$

- A.  $64^\circ$ .
- B.  $69^\circ$ .
- C.  $72^\circ$ .
- D.  $78^\circ$ .

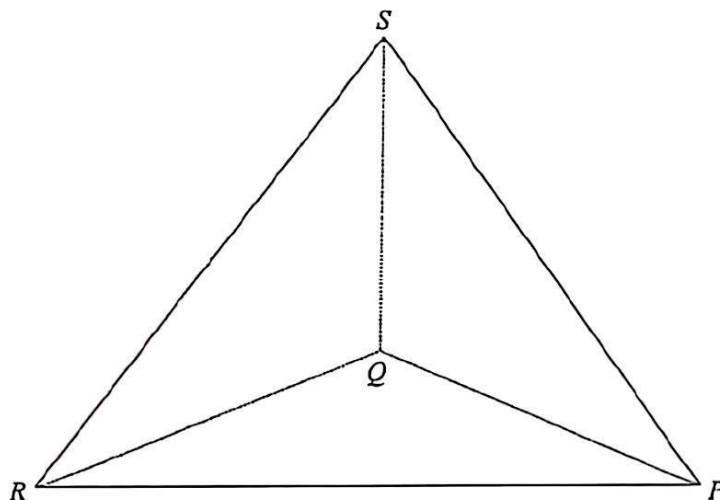


39. For  $0^\circ < \theta \leq 360^\circ$ , how many roots does the equation  $4\cos^2\theta - 3\cos\theta - 1 = 0$  have?

- A. 3
- B. 4
- C. 5
- D. 6

40. In the figure, the base  $PQR$  of the tetrahedron  $PQRS$  lies on the horizontal ground. It is given that  $Q$  is vertically below  $S$ . If  $\angle PQR = 90^\circ$ ,  $\angle QPS = 30^\circ$  and  $\angle QRS = 45^\circ$ , then  $\cos \angle PRS =$

- A.  $\frac{1}{2}$  .  
 B.  $\frac{3}{4}$  .  
 C.  $\frac{\sqrt{2}}{4}$  .  
 D.  $\frac{\sqrt{3}}{6}$  .



41. Let  $G$ ,  $H$ ,  $I$  and  $J$  be the centroid, the orthocentre, the in-centre and the circumcentre of  $\triangle PQR$  respectively. If  $\angle PQR = \angle PRQ = 22^\circ$ , which of the following are true?

- I.  $G$  lies inside  $\triangle PQR$  .  
 II.  $H$  lies outside  $\triangle PQR$  .  
 III.  $I$ ,  $J$  and  $Q$  are collinear.

- A. I and II only  
 B. I and III only  
 C. II and III only  
 D. I, II and III
42. A queue is formed by 2 managers and 7 officers. If no managers are next to each other, how many different queues can be formed?
- A. 80 640  
 B. 141 120  
 C. 282 240  
 D. 362 880

43. There are three questions in a test. The probabilities that a student answers the first question correctly, the second question correctly and the third question correctly are 0.6, 0.7 and 0.8 respectively. Find the probability that the student answers at least 1 question correctly in this test.

A. 0.024  
B. 0.188  
C. 0.812  
D. 0.976

44. In an examination, the scores (in marks) of the candidates are as follows:

39    10    13    16    17    19    25    26    28    30    30    32

Which of the following is/are true?

- I. The median of the examination scores of the candidates is 22 marks.  
II. The standard score of each candidate in the examination is lower than 2.  
III. The standard deviation of the examination scores of the candidates exceeds 8 marks.

A. I only  
B. II only  
C. I and III only  
D. II and III only

45. If the variance of the seven numbers  $x_1$ ,  $x_2$ ,  $x_3$ ,  $x_4$ ,  $x_5$ ,  $x_6$  and  $x_7$  is 16, then the standard deviation of the seven numbers  $9x_1 - 5$ ,  $9x_2 - 5$ ,  $9x_3 - 5$ ,  $9x_4 - 5$ ,  $9x_5 - 5$ ,  $9x_6 - 5$  and  $9x_7 - 5$  is

A. 31.  
B. 36.  
C. 139.  
D. 144.

**END OF PAPER**